

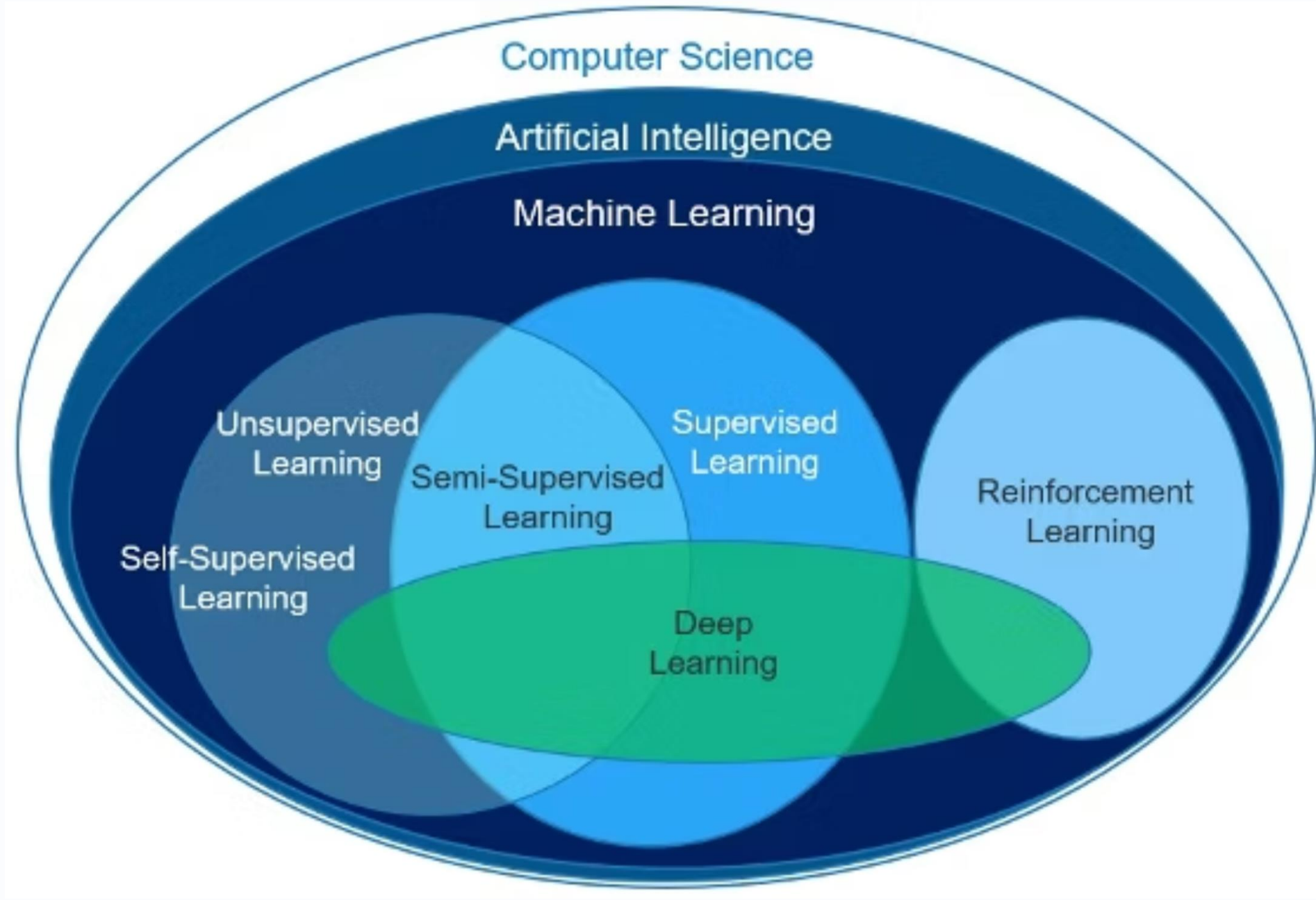
DAI 1000

Digital and AI Literacy

L5 - Machine Learning Essential I

Content

- **Key Concepts**
- Key Steps in Machine Learning
 - Stage 0: Prepare the dataset
 - Stage 1: Design the Model
 - Stage 2: Identify the key values (e.g., Parameters and Hyperparameters)
 - Stage 3: Evaluate the model





Input

- Stock Data
- Customer Transaction Data
- Streaming Data
- Email text



Machine Learning Techniques

- Regression
- Clustering
- Association Rule
- Classification



Output

- Stock Price Prediction
- Market Segmentation
- Recommendations Systems
- Spam Detection

Machine Learning

Computers learn from data and make predictions without being explicitly programmed



Learning

The machine is fed data (text, images, or sensor signals) and extracts patterns through algorithms



Inference

Once trained, the machine takes new input (data) and produces output (e.g., classification or prediction)

Machine Learning Examples

Predicting house prices



Learning

You feed it data from past house sales (input: features and their corresponding prices)

The system learns the underlying relationship, identifying a function: $Price = f(\text{Square Footage}, \text{Number of Rooms}, \text{Location})$.



Inference

New Inputs (x): Square footage, number of rooms, location, age of a new house

→ Output (y): The estimated price of the new house

Machine Learning Examples

Image Recognition

Learning



Provide labeled images (e.g., Dog and Cat images)

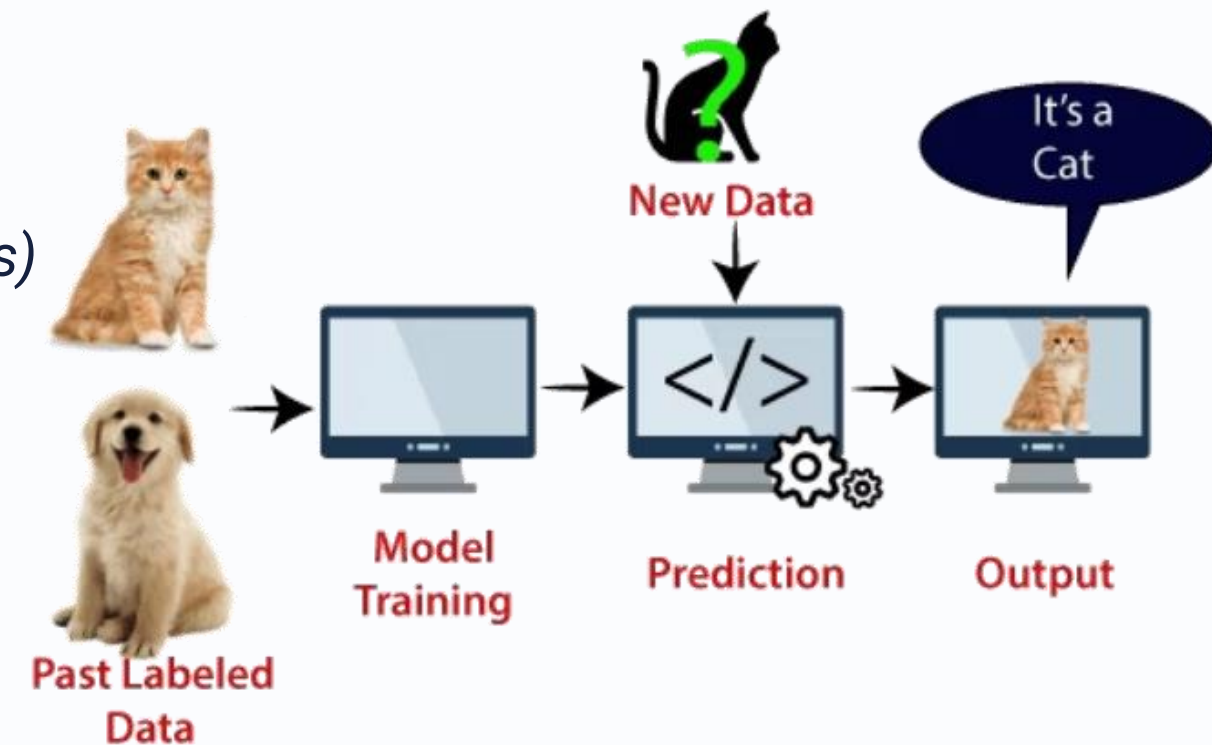
The machine learns a function: $Object = f(pixel\ values)$

Inference



New Inputs (x): Unlabeled new images

→ Output (y): Object label (cat or dog)



Machine Learning

— — Finding a Function

Traditional Programming: Human explicitly define this function

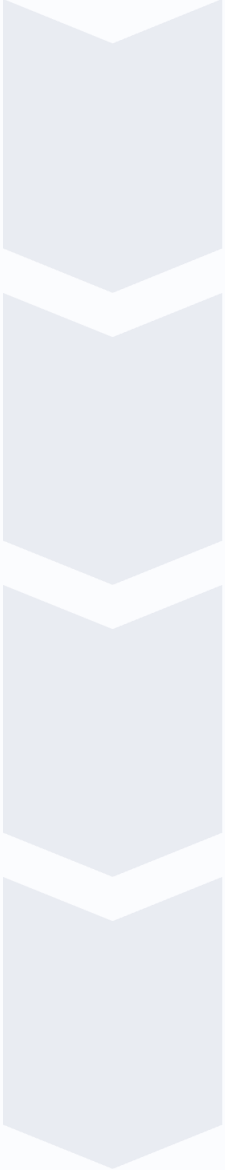
Machine Learning: From training data, finding a function, $f(x)$, that maps a set of inputs (x) to a desired output (y).

E.g., in $y = ax + b$, finding the values of a and b

Content

- Key Concepts
- **Key Steps in Machine Learning**
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Machine Learning Process Overview



Stage 0: Prepare the dataset

Collect, preprocess, and organize data for training

Stage 1: Design the Model

Select appropriate architecture for your task

Stage 2: Identify the key values

Train the model using loss functions and optimization

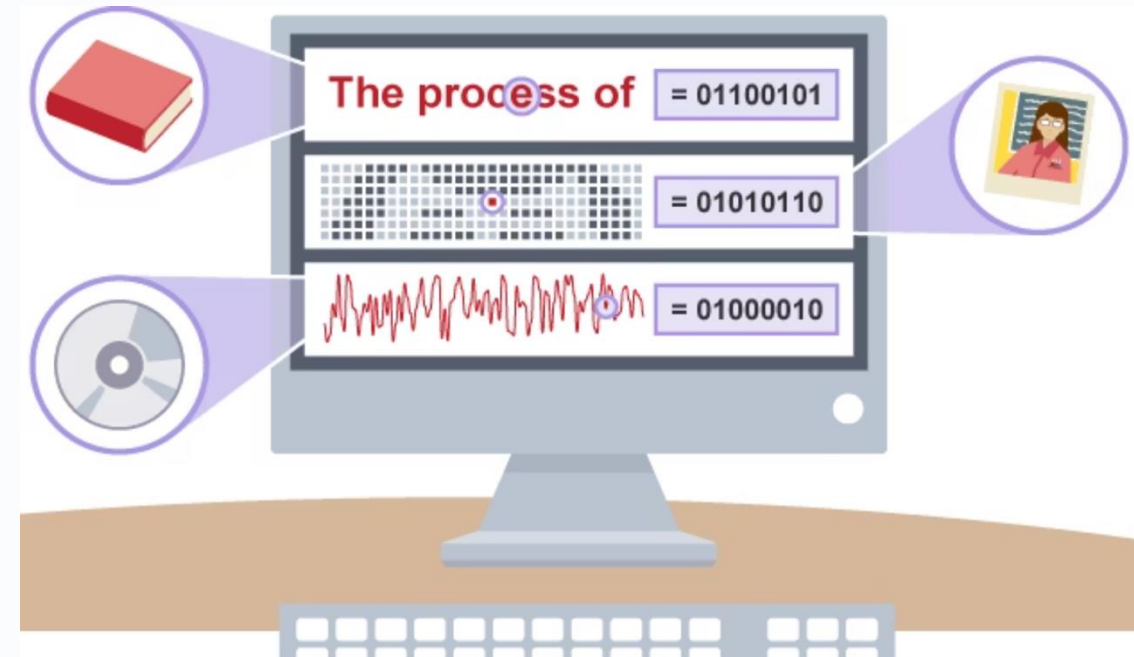
Stage 3: Evaluate the model

Test performance and refine as needed

Stage 0: Prepare the Dataset

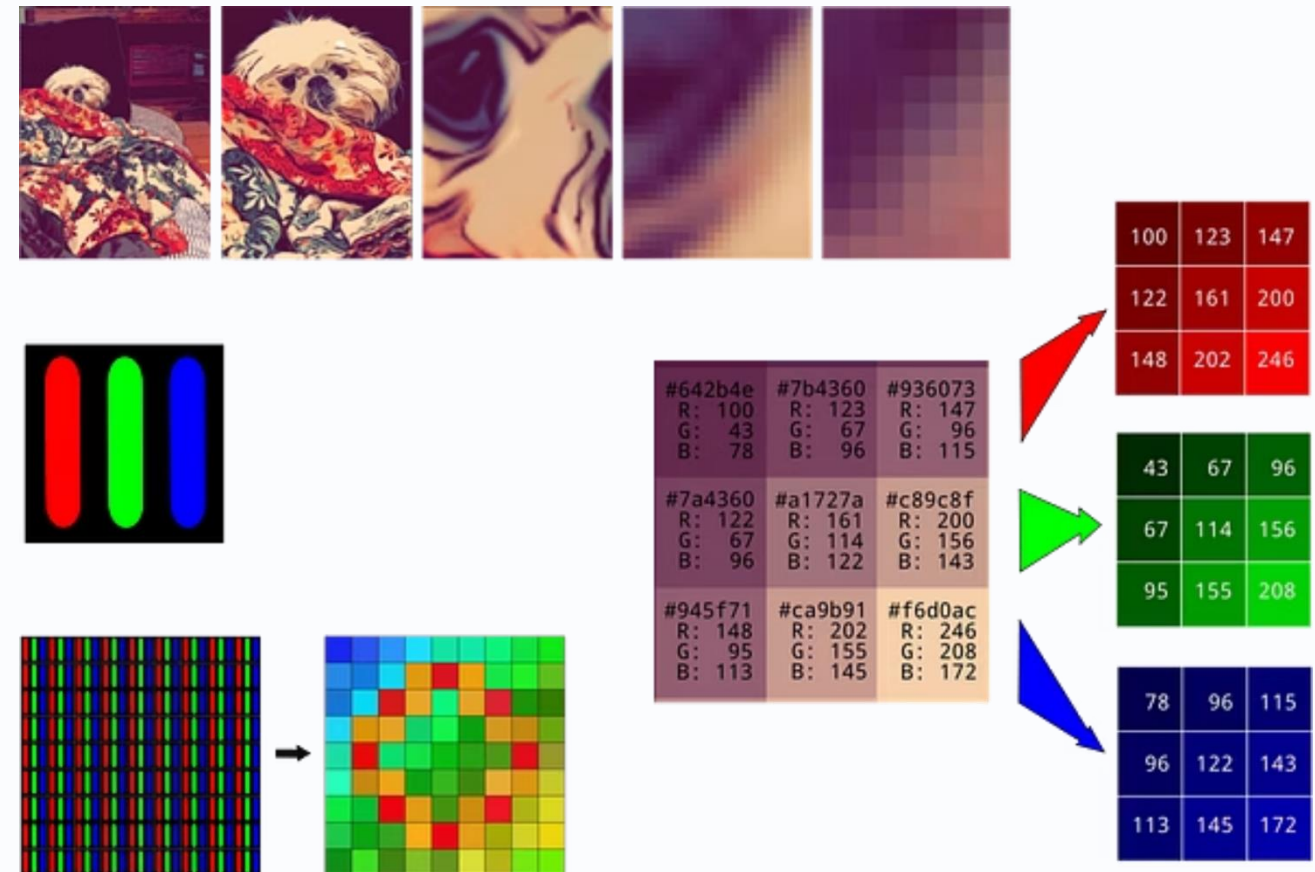
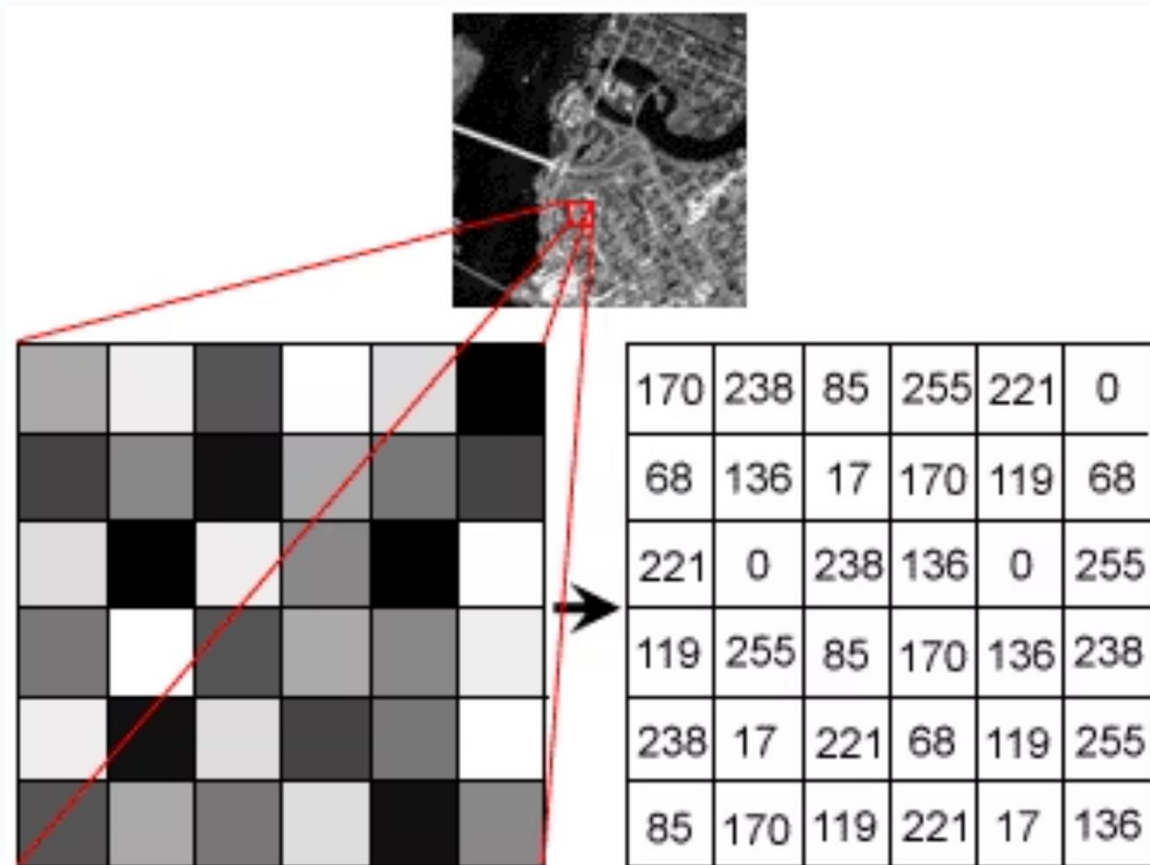
Data can come in many forms:

- Numerical tables
- Text
- Images
- Audio
- Video

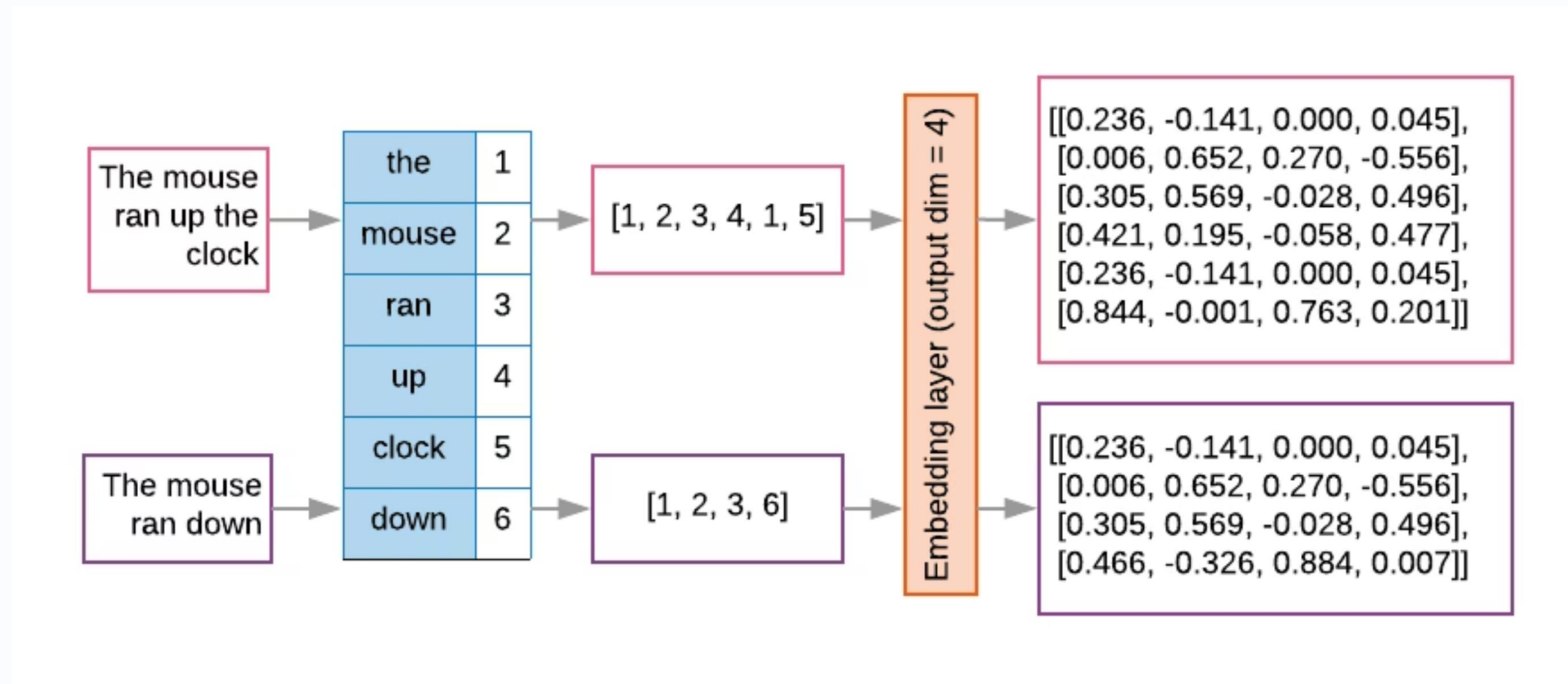


All of these data types can be converted into numerical representations so they can be processed by machine learning algorithms.

Images are converted into pixel intensity values



Text is tokenized and embedded into vectors



Data Preparation Process

Data Collection

Gather large amounts of relevant **data** from various sources

Data Cleaning

- **Removing duplicates** to ensure each entry is unique and relevant
- **Correcting formatting inconsistencies** by standardizing formats
- **Handling missing values** using mean/mode imputation, deletion, or predictive models

Data Annotation

Labels are required, especially in supervised learning.

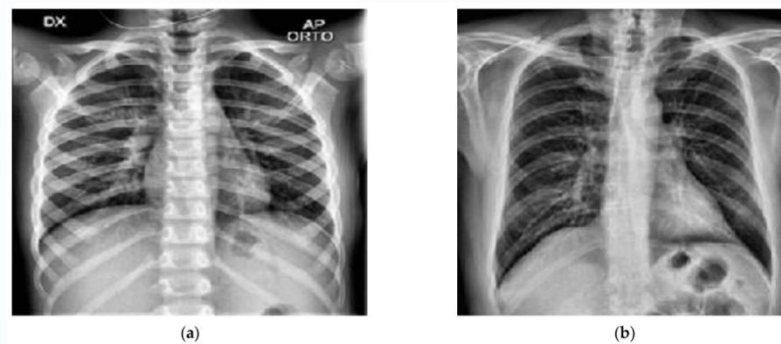
Human annotators—such as doctors labeling medical images—provide the ground truth used to train the model.

This process is **manual, time-consuming, and expensive**.

Real-World Data Preparation Examples

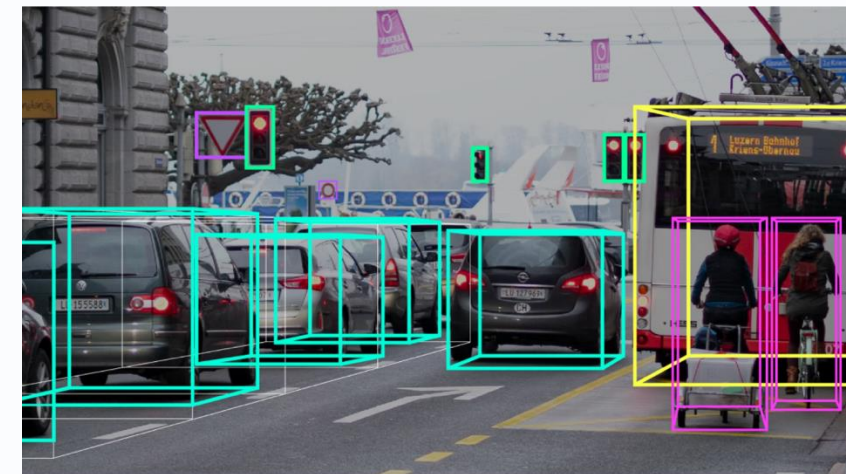
COVID-19 Chest Scan Classification

- **Task:** Training AI to detect diseases from chest X-ray images
- **Dataset:** Public medical datasets with train/val/test splits
- **Label:** Diagnosis provided by doctors (e.g., COVID-positive or negative)



3D Street Scene Annotation

- **Task:** Recognize and segment objects (e.g., cars, trees, road signs) in 3D urban environments
- **Label:** Pixel-wise or bounding box annotations created manually
- **Challenge:** Extremely labor-intensive annotation process involving many frames and object classes



Data Splitting

Training Set

Train the machine learning model.

The model learns patterns and relationships to optimize its internal parameters.



Validation Set

Used during the model development phase to tune hyperparameters and select the best model configuration.

It provides an unbiased evaluation during hyperparameter tuning, preventing the model from overfitting to the training data.

Test Set

To evaluate the performance of the fully trained and optimized model.

It provides an independent and unbiased assessment of the model's generalization capabilities on new, real-world data.

Stage 1: Design the Model

In this stage, you define the model architecture based on the task. The model type determines how the function will map inputs to outputs.

Map of ML

Paradigm →

Tasks →

Algorithms/ Models →

Supervised Learning

Classification

Regression

KNN, linear models, DNNs ...

Semi/Unsupervised Learning

Clustering

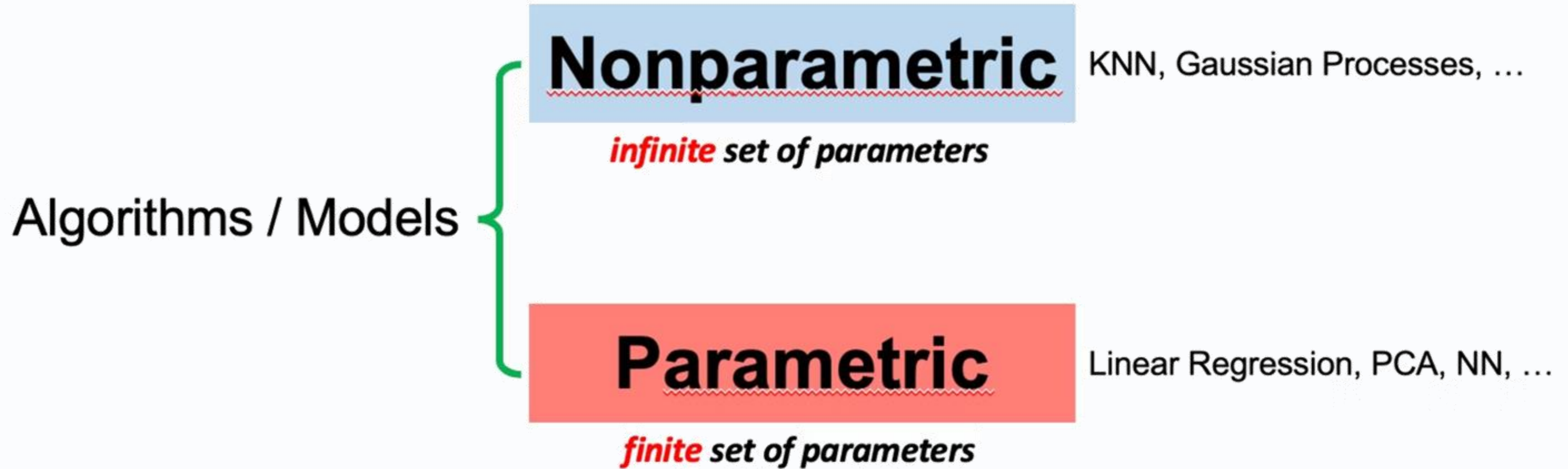
*Semi-sup
Learning*

*Generative
Model*

....

K-means, PCA, DNNs ...

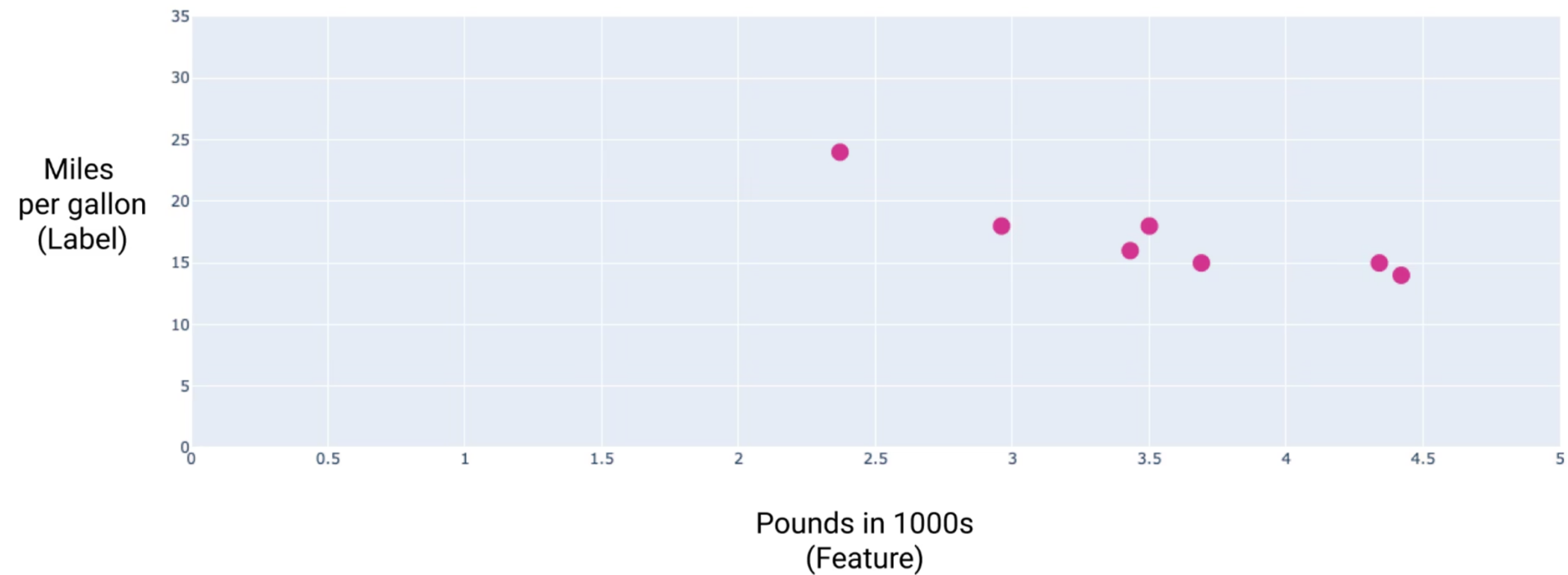
Reinforcement Learning



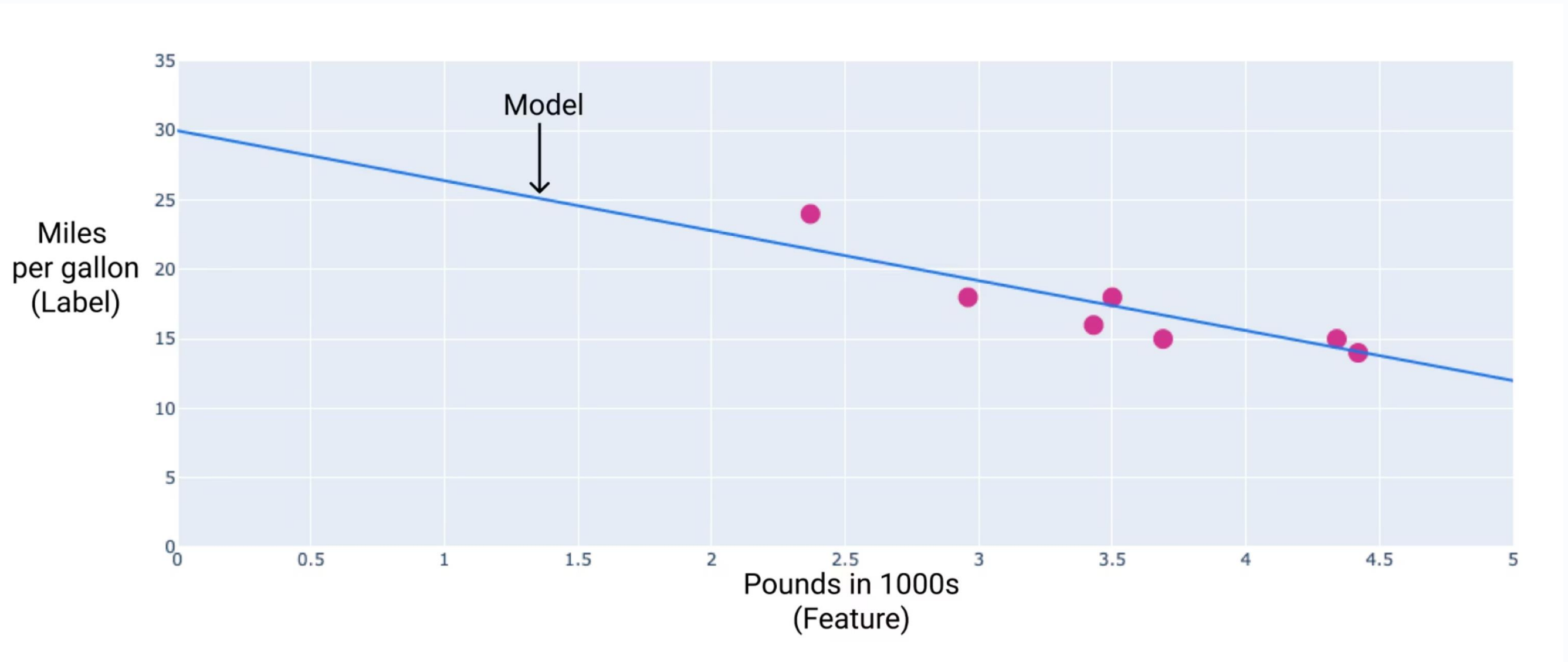
Task Examples:

Model	Description	Typical Uses
Linear Regression (线性回归)	Fitting a line to predict continuous values	House price prediction, income forecasting
Logistic Regression (逻辑回归)	Predicting classification probabilities (binary)	Disease diagnosis, purchase prediction
KNN (K-最近邻)	Looking at K nearby samples	Image classification, recommendation systems

Linear Regression



Car heaviness (in pounds) versus miles per gallon rating



A best fit line drawn through the data from the previous figure.

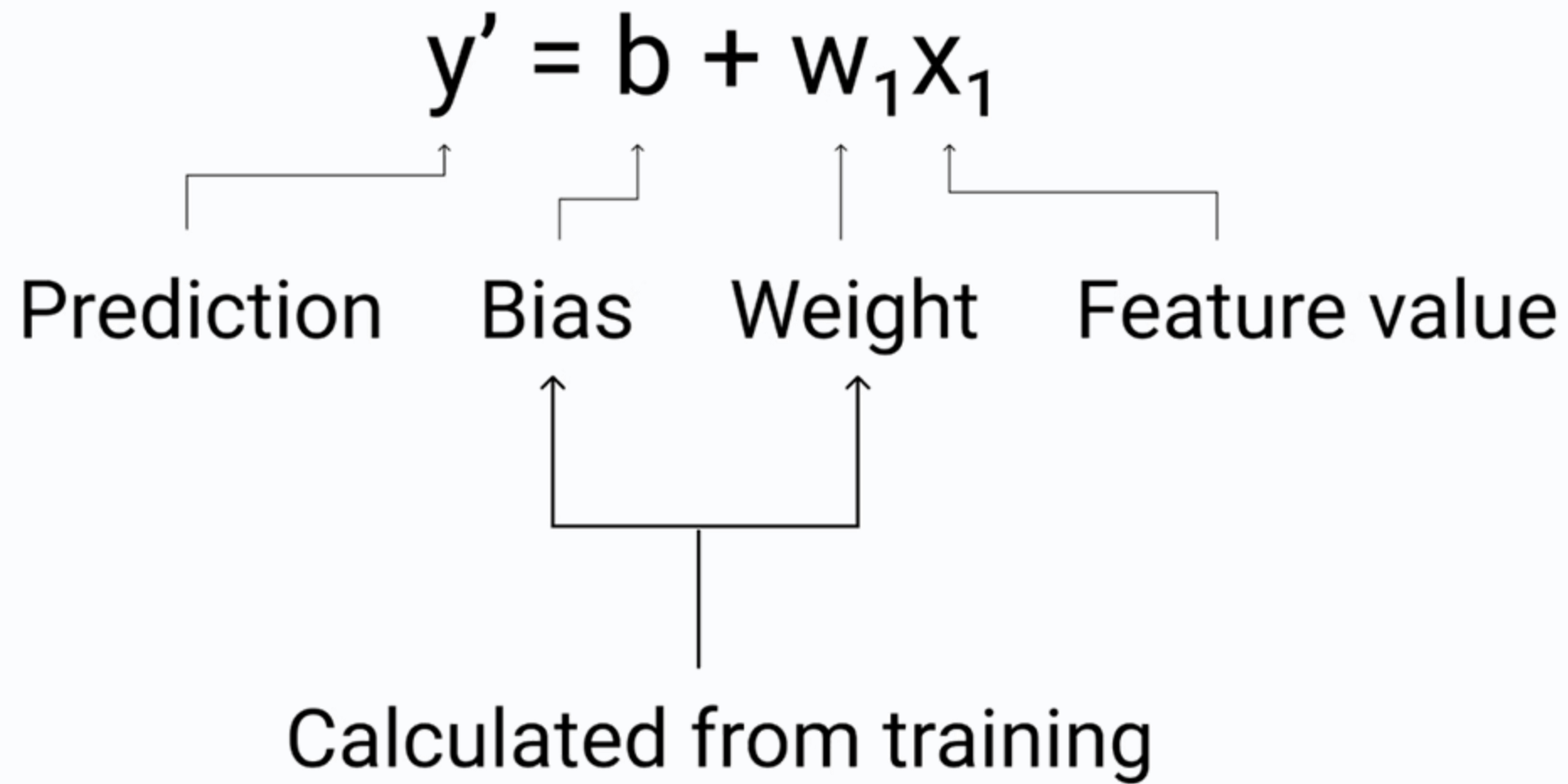
In algebraic terms, the model would be defined as **$y=mx+b$** , where

- y is miles per gallon—the value we want to predict.
- m is the slope of the line.
- x is pounds—our input value.
- b is the y -intercept.

Stage 2: Identify the key values

This stage is about training the model to learn the best values (e.g., **parameters** and **hyperparameters**)

Parameters: b and w₁



Parameters: b, w1, w2, w3, w4, w5

$$y' = b + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + w_5 x_5$$

The diagram illustrates the mapping of car features to weights in a linear regression model. The equation $y' = b + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + w_5 x_5$ is shown at the top. Below it, five features are listed: Pounds, Displacement, Acceleration, Number of cylinders, and Horsepower. Arrows connect these features to their respective weights in the equation: Pounds to w_1 , Displacement to w_2 , Acceleration to w_3 , Number of cylinders to w_4 , and Horsepower to w_5 . Each weight w_i and its corresponding feature x_i are enclosed in a dashed box.

Pounds Displacement Acceleration Number of cylinders Horsepower

What Parameter Values are the best?

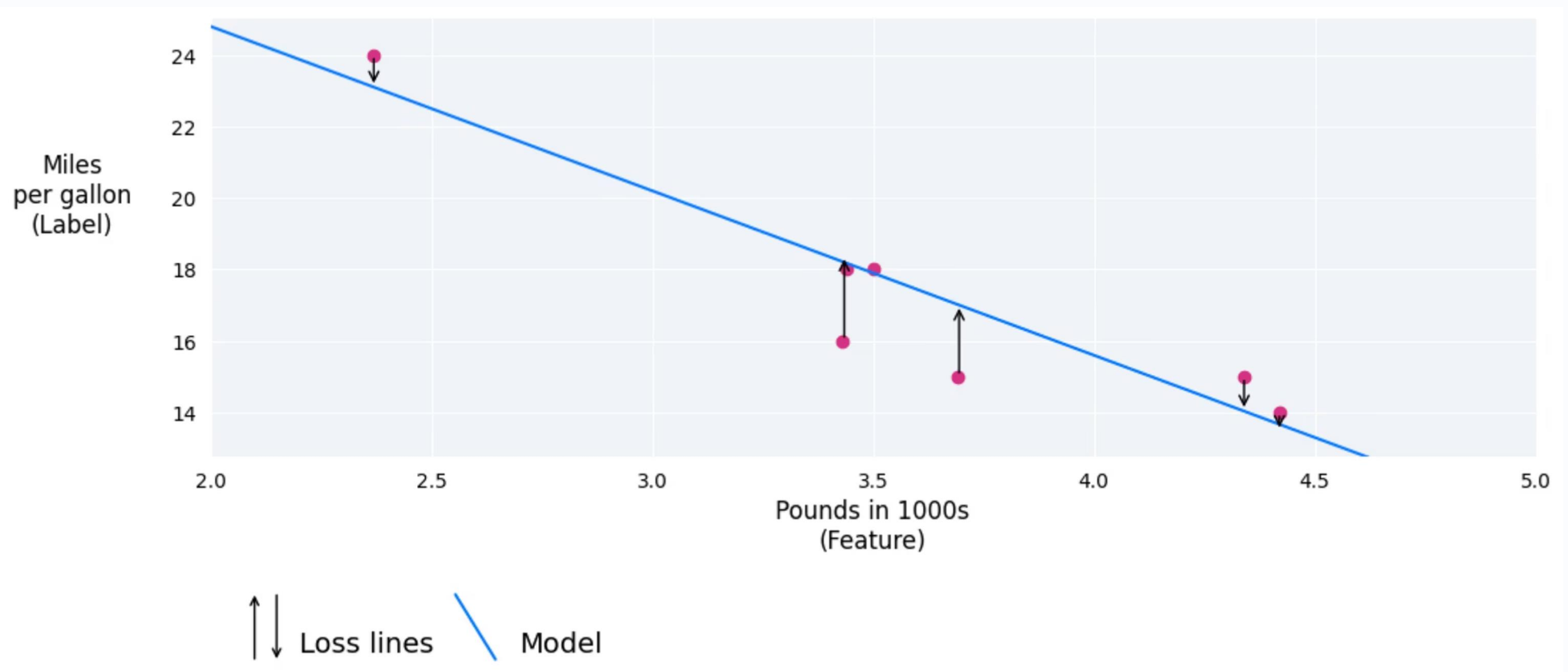
Need to find the parameters to **minimize the loss**.

Loss: measures the distance between the model's predictions and the actual labels

Example: For regression, a common one is Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where y_i is the true value and $\hat{y}_i$ is the predicted value. A lower loss means better performance on the training data.



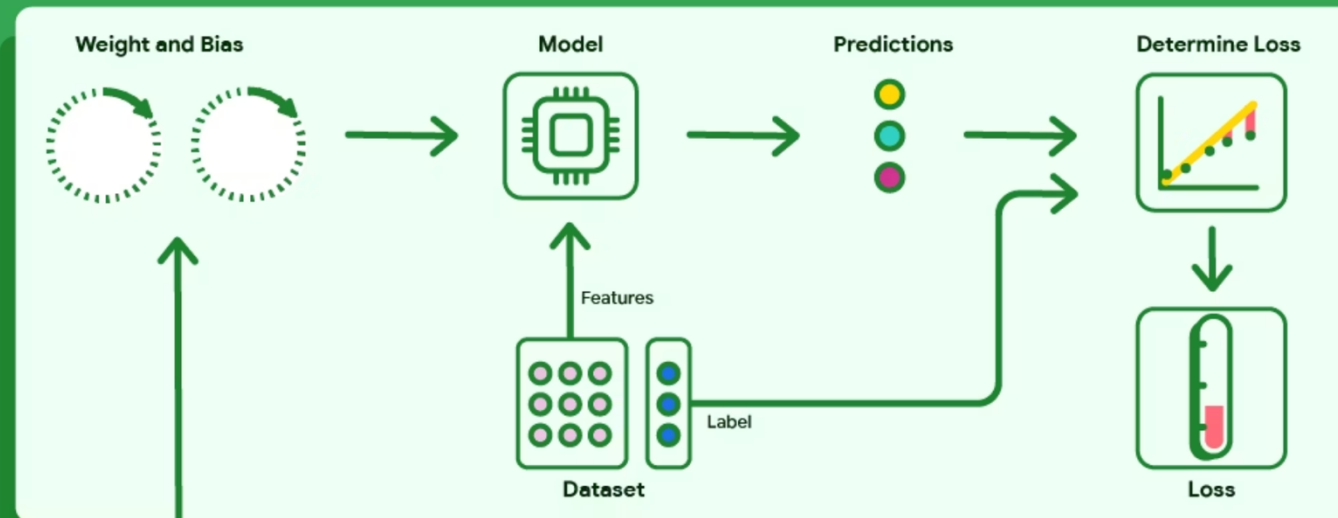
How to find the parameter values that minimize the LOSS?

Gradient descent

A mathematical technique that iteratively finds the weights and bias that produce the model with the lowest loss.

<https://www.bilibili.com/video/BV1HBJ9zRE6N/>

1 Calculate loss



4

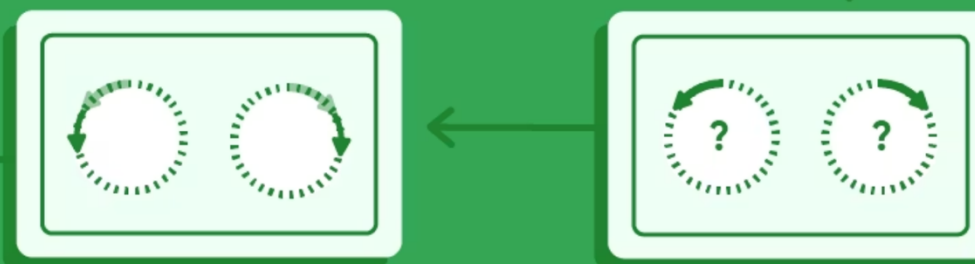
Repeat the process until loss can't be reduced

3

Move a small amount in the direction that reduces loss

2

Determine the direction to move the weights and bias



Hyperparameters

Hyperparameters are values that human control.

- Learning rate
- Batch size
- Epochs

Parameters are values that the model calculates during training.

- Learning rate

Learning rate is a floating point number you set that influences how quickly the model converges. If the learning rate is too low, the model can take a long time to converge.

- Batch size

Batch size is a hyperparameter that refers to the number of examples the model processes before updating its weights and bias.

- Epochs

An epoch means that the model has processed every example in the training set once.

Full batch: an entire dataset

	Feature	Label
0		
•		
•		
•		
•		
500		
•		
•		
•		
•		
1000		

Single batch: a mini batch

	Feature	Label
1		
•		
•		
•		
100		

One epoch: a full batch composed of ten mini batches

	Feature	Label
1		
•		
•		
100		
101		
•		
•		
200		
201		
•		
•		
300		
301		
•		
•		
400		
401		
•		
•		
500		

	Feature	Label
501		
•		
•		
600		
601		
•		
•		
700		
701		
•		
•		
800		
801		
•		
•		
900		
901		
•		
•		
1000		

Example: Given a training set with 1,000 examples and a mini-batch size of 100 examples, it will take the model 10 iterations to complete one epoch.

Optimization: Finding the Best Function

Least Loss

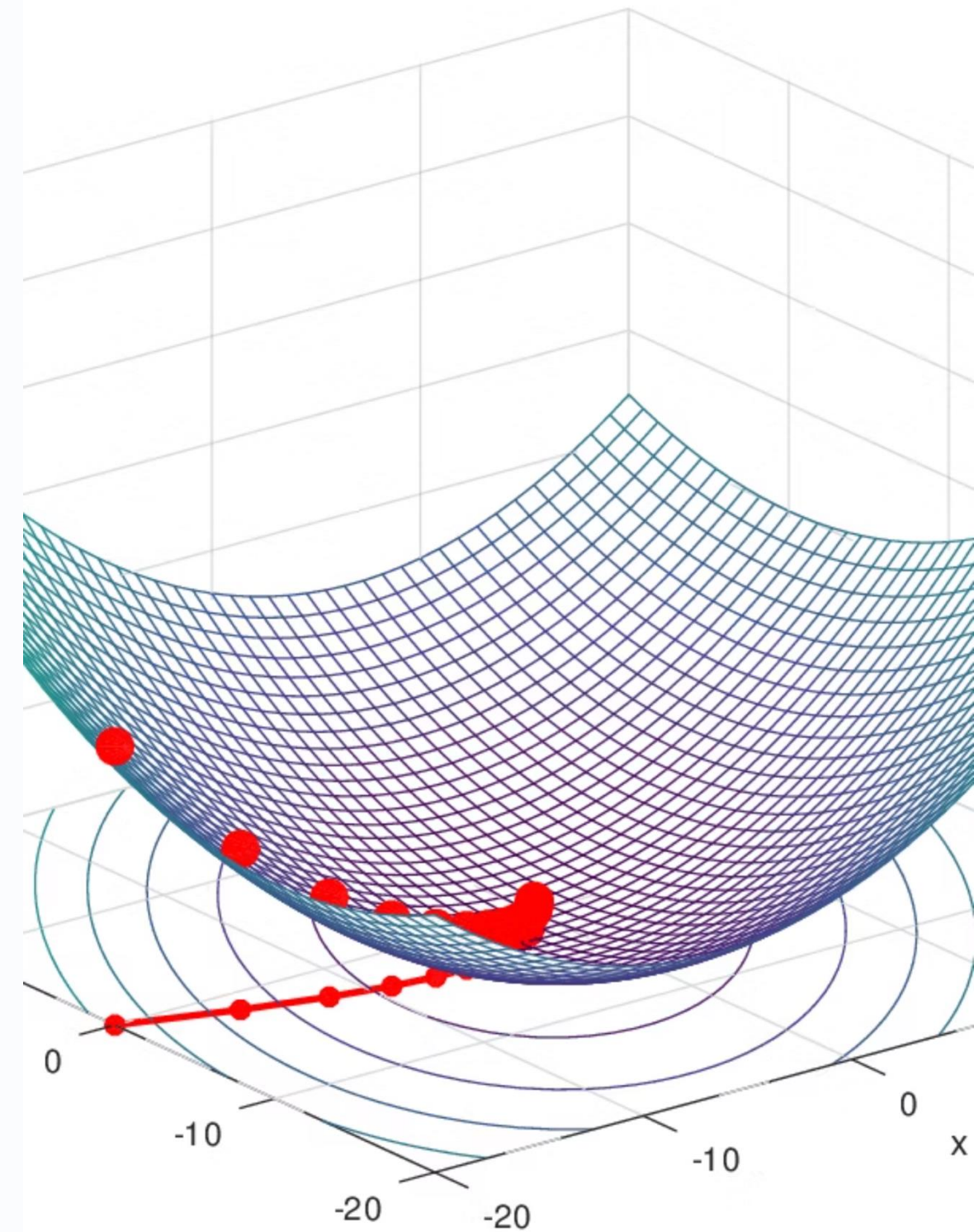
Gradient Descent technique minimizes the loss function to find optimal parameters

Faster Model Running

Optimized algorithms improve computational efficiency

Less Sensitivity

Reduced dependency on hyperparameter tuning for more stable performance



Stage 3: Evaluate the model

Even if a model achieves low training loss, it may not generalize well to new data. This stage assesses generalization.

Use Test Set

Evaluate performance on unseen data to measure true effectiveness

Apply Metrics

For regression: Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2)

Compare Results

Good models have low loss on training and test sets. Large **performance drops on test sets indicate overfitting**

i **Poor Generalization Example:** A house price prediction model with low MSE on training data but high MSE on test data

Sources

- https://medium.com/@ketan.patel_46870/text-to-numeric-representation-in-nlp-a-beginner-friendly-guide-9e68c8f8d07c
- <https://www.kdnuggets.com/2020/01/convert-picture-numbers.html>
- <https://www.bbc.co.uk/bitesize/guides/zpfdwmn/revision/1>
- <https://developers.google.com/machine-learning/guides/text-classification/step-3>
- <https://www.youtube.com/watch?v=4RixMPF4xis>
- https://developers.google.com/machine-learning/crash-course/linear-regression?_gl=1*1u9fwqw*_up*MQ..*_ga*MzI2ODI5NzU5LjE3NTg3MDAyMTI.*_ga_SM8HXJ53K2*czE3NTg3MDAyMTIkbzEkZzAkdDE3NTg3MDAyMTIkajYwJGwwJGgw

Questions?

Course Feedback

